📘 Project Logbook

*Week 6: A Pathfinding Algorithm and AR Navigation**

Overview:
The sixth week marked a turning point in merging backend data with real-time AR guidance. The objective was to use the A* algorithm to compute indoor routes and project them using arrows in the user's physical space.

Progress:

- Developed A* algorithm in Java, defining nodes for facilities and entry/exit points.
- Modeled the test station layout as a graph with weighted edges.
- Connected AR anchors to graph nodes for real-time path traversal.
- Enabled the app to fetch user location, compute shortest path, and return directional steps.
- Rendered AR arrows between consecutive path nodes.
- Implemented user re-calibration in case of deviation from computed path.
- Optimized the rendering speed to ensure minimal latency during path updates.
- Logged anchor-to-anchor distance accuracy to ensure correct directionality.

Summary:
The application now successfully offers guided navigation using computed shortest paths in AR view. This is the first version of a dynamic, spatially aware path guidance system within a mobile app.

Next Week and Beyond:
With major components working, we now aim to conduct structured user testing to identify usability flaws and performance gaps, especially in busy or low-light environments.